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14 November 1979
SUPERSEDING
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29 July 1974

MILITARY SPECIFICATION

DETECTOR UNIT, CHEMICAL AGENT AUTOMATIC ALARM: ABCA-M43

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes requirements for a portable, automatic VG agent detector.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel, and Seals.
PPP-B-601 - Boxes, Wood, Cleated Plywood.
PPP-B-621 - Box, Wood, Nailed and Lock Cornered.

FSC 6665

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-252	-	Classification of Visual and Mechanical Defects For Equipment, Electronic, Wired, and Other Devices.
MIL-STD-781	-	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution.
MIL-STD-810	-	Environmental Test Methods.
MIL-STD-1188	-	Commercial Packaging of Supplies and Equipment.

DRAWINGS AND TECHNICAL DATA PARTS LIST (TDPL)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

TDPL 5-15-4400	-	Detector Unit, Chemical Agent Automatic Alarm: ABCA-M43.
C5-15-5479	-	Buffer Solution.
D5-15-6567	-	Packaging of Detector Unit, ABCA-M43.
D136-41-1520	-	Portable Gas Generator, Q219.
D136-45-245	-	Tester, Functional, Detector Unit, Subassembly, Q200.

PUBLICATIONS

DEPARTMENT OF THE ARMY TECHNICAL MANUAL

TM3-6665-225-12	-	Alarm, Chemical Agent, Automatic: Portable, ABCA-M8.
TM3-6665-302-34	-	Detector Unit, Chemical Agent, Automatic Alarm, M43 and Alarm Unit, Chemical Agent, Automatic Alarm, M42.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on TDPL 5-15-4400 or on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings and in this specification.

3.1.2 Components. All components of the detector unit shall conform to the specifications and drawings listed on TDPL 5-15-4400 and subsidiary drawings.

3.2 Manufacture and assembly. The components shall be manufactured and assembled as specified on the subsidiary drawings listed on Drawing D5-15-4400.

3.3 Air and liquid system.

3.3.1 Airflow at air inlet. The airflow at the air inlet of the completely assembled detector, with the air filter in place, shall be 1.1 (+0.4 or -0.0) liter per minute with 24.0 ± 1.0 volt direct current (vdc) applied at the detector's 24 VDC INPUT when tested as specified in 4.4.4.1.

3.3.2 Resistance to airflow. The resistance to airflow of the several sections of the air and liquid system shall be as required below when tested as specified in 4.4.4.2.

(a) Detector air inlet to the chassis seal at the air pump inlet. At an airflow rate of 1.1 ± 0.2 liter per minute, with a simulated detector cell in place, the resistance to airflow of the air system from the detector air inlet to the chassis seal at the air pump inlet shall not exceed a pressure drop of 3.0 inches (7.62 centimeters (cm)) of water.

(b) Chassis seal at the air pump outlet to the detector air outlet. At an airflow rate of 1.4 (+0.1 -0.0) liter per minute, the resistance to airflow of the air system from the chassis seal at the air pump outlet to the detector air outlet shall not exceed a pressure drop of 1.0 inch (2.54 centimeters (cm)) of water.

(c) Weight assembly in liquid reservoir to the chassis seal at the liquid pump inlet. At an airflow rate of 0.25 ± 0.05 liter per mirute, the resistance to airflow of the liquid system from the weight assembly in the liquid reservoir to the chassis seal at the liquid pump inlet shall not exceed a pressure drop of 10.0 inches (25.40 cm) of water.

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(d) Chassis seal at the liquid pump outlet to the chassis seal at the detector cell liquid inlet. At an airflow rate of 0.25 ± 0.05 liter per minute, the resistance to airflow of the liquid system from the chassis seal at the liquid pump outlet to the chassis seal at the detector cell liquid inlet shall not exceed a pressure drop of 5.0 inches (12.70 cm) of water.

3.3.3 Air leakage at the detector's air filter seal. Air leakage around or past the detector's air filter seal shall not exceed a rate of 0.7 cubic centimeter per minute when a blank air filter is installed in place of the detector's air filter, and the downstream side of the air filter is at a pressure of 6.0 ± 0.4 inch (15.24 ± 1.02 cm) of water less than the upstream side when tested as specified in 4.4.4.3. For the purpose of this requirement, a blank air filter is defined as an air filter made from the same material and having the same interface dimensions as the detector's air filter (Drawing C5-15-4720) but is impermeable to the passage of air.

3.3.4 Air leakage of the combined air and liquid system. The static air leakage rate of the combined air and liquid system shall not exceed 1.9 cubic centimeters per minute with the internal air pressure of the combined air and liquid system at a pressure 49.0 ± 1.0 inch (124.46 ± 2.54 cm) of water greater than the external pressure when tested as specified in 4.4.4.4.

3.3.5 Air leakage of the liquid line. The static air leakage rate of the liquid line shall not exceed 0.5 cubic centimeter per minute with the internal air pressure of the liquid line at a pressure 49.0 ± 1.0 (124.46 ± 2.54 cm) inch of water greater than the external pressure when tested as specified in 4.4.4.5.

3.4 Temperature control system.

3.4.1 Chassis heater circuit current. With the chassis heater circuit control thermostat (1A2A2S1) cooled below its operating point, and 24.0 ± 1.0 vdc applied at the detector's 24 VDC INPUT, the chassis heater circuit current shall be 87.0 ± 10.0 milliamperes when tested as specified in 4.4.4.6.

3.4.2 Interior thermostat operation. With the interior thermostat (1A2A2S2) at room temperature, the resistance between 1A2A2A5J3-C and 1A2A2A5J1-W shall not exceed 1.0 ohm. With the interior thermostat cooled below its operating point, the resistance between 1A2A2A5J3-C and 1A2A2A5J1-W shall be at least 1.0 megohm when tested as specified in 4.4.4.7.

3.4.3 Outlet heater circuit current. With the outlet heater circuit control thermostat (1A2A5A1S1) cooled below its operating point, and 24.0 ± 1.0 vdc applied at the detector's 24 VDC INPUT, the outlet heater current shall be 39.0 ± 5.0 milliamperes when tested as specified in 4.4.4.8.

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3.4.4 Inlet heater current and operating temperature. With an airflow rate at the detector air inlet of 1.1 + 0.4 -0.0 liter per minute, the air entering the detector air inlet at a temperature of 30.0° + 15° -5.0°F (-1.11° + 8.33° - 2.78°C) and 24.0 ± 1.0 vdc applied at the detectors 24 vdc input, the current through the inlet heater shall be 145 ± 15 milliamperes when the heater is energized, when tested as specified in 4.4.4.9. With 28.0 ± 9.0 vdc power source applied at the detectors 24 vdc input, the lowest temperature measured immediately below the bottom filter of the air filter shall be 92.5° ± 2.5°F (33.61° ± 1.39°C) when tested at the end of the 12 hour ambient temperature stability run as specified in 4.4.4.17.

3.5 Electrical functioning.

3.5.1 Simulated alarm. With 24.0 ± 1.0 vdc applied at the detector's 24 VDC INPUT, and the detector in the alarm mode of operation, the detector shall meet the following requirements when tested as specified in 4.4.4.10:

(a) A voltage within 2.5 volts of the applied voltage shall appear at the REMOTE terminals.

(b) The pump unit motor shall stop running.

(c) The inlet heater shall stay off.

(d) The horn shall sound with a varying frequency.

(e) The horn sound output level shall decrease as the HORN VOL control is rotated from fully clockwise (cw) to full counterclockwise (ccw) position.

3.5.2 Meter zero adjustment circuit. With 24.0 ± 1.0 vdc applied at detector's 24 VDC INPUT, operation of the ZERO ADJUST control shall cause the meter pointer to move to the limit of its travel on the left end of the meter scale and to at least 20 on the right side of the zero zone when tested as specified in 4.4.4.11.

3.5.3 Meter accuracy. With 24.0 ± 1.0 vdc applied at the detector's 24 VDC INPUT, operation of the BATTERY TEST switch shall cause the meter to indicate a voltage within 2.0 volts of the applied voltage when tested as specified in 4.4.4.12.

3.5.4 Detector operating current. With 24.0 ± 1.0 vdc applied at the detector's 24 VDC INPUT, and the detector operating with 1.1 (+ 0.4 or - 0.0) liter per minute airflow, the current required by the operating detector (exclusive of all heaters) shall not exceed 85.0 milliamperes when tested as specified in 4.4.4.13.

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3.5.5 Resistances. When tested as specified in 4.4.4.14, the resistances measured between the points listed in table I shall be as specified in table I.

3.6 Fast priming mechanism. When tested as specified in 4.4.4.15 the fast priming mechanism shall demonstrate these capabilities:

(a) As the shaft of the PRIME crank is pulled into its uppermost position the shouldered pin (Drawing C5-15-4648) in the tube guide shall be displaced from its normal operating position at least 0.04 inch (1.02 mm), measured along the surface of the pump housing (Drawing E5-15-4601).

(b) As the shaft of the PRIME crank is pushed into its center position, the roller (Drawing B5-15-4533) in the bell crank assembly shall move away from the shouldered pin. With the shaft of the PRIME crank fully detented into its center position, the roller shall not touch the shouldered pin.

(c) As the shaft of the PRIME crank is pushed into its lowermost position, the end of the shaft shall come into contact with the face of the fin housing (Drawing C5-15-4608). With the PRIME crank shaft held in contact with the fin housing, and the PRIME crank rotated in the direction indicated on the case top, the pin in the end of the shaft shall engage the fast prime fin (Drawing C5-15-4618) and rotate the fin housing. When the PRIME crank is released, the shaft shall return to its center position.

3.7 Detector leakage. The detector assembly shall not leak when submerged in water and pressurized internally with filtered air at 50 ± 2 inches (127 ± 5.08 cm) of water greater than the ambient pressure for one minute when tested as specified in 4.4.4.16.

3.8 All equipment screening test.

3.8.1 Vibration. When tested as specified in 4.4.4.17.1 the detector shall exhibit no physical deterioration and shall be capable of being operated as prescribed by TM3-6665-225-12.

3.8.2 Stability. When tested as specified in 4.4.4.17.2 the detector shall not false alarm (see 6.5.7), malfunction (see 6.5.8), exhibit drift (see 6.5.1) in excess of 1.0 volt peak-to-peak or noise and transients (see 6.5.2) in excess of 200 millivolts peak-to-peak, during a monitoring period of at least 12 hours. A specialized form of transient, a minus spike which would cause the detector meter to deflect downscale, may occur during the monitoring period provided that the magnitude of the minus spike does not exceed 300 millivolts. At the conclusion of the test, the detector shall have proper airflow at the air inlet and shall respond to the simulant and solution as described in TM3-6665-225-12.

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Table I. Resistance between points

From	To	Resistance (ohms)
1A2A5A1J1-A	1A2A2A4P4	Less than 1.0
1A2A5A1J1-C	1A2A2A5J3-D	Greater than 1000
1A2A5A1J1-D	1A2A5A2-E1	Less than 1.0
1A2A5A1J1-C	1A2A2A5J1-H	Less than 1.0
1A2A5A1J1-B	1A2A2A5J1-W	Less than 1.0
1A2A5A1J1-E	1A2A2A5J3-A	Less than 1.0
1A2A5A2-E2	1A2A2A5J1-H	Less than 1.0
1A2A2A5J3-A	1A2A2A5J1-T	Less than 1.0
1A2A2A5J3-D	1A2A2A5J1-J	Less than 1.0
1A2A2A5J1-D	1A2A2A5J1-K	Greater than 1000; less than 1.0 with ZERO knob depressed
1A2A5A2-E1	1A2A2A5J1-Q	Less than 1.0
1A2A2A5J3-B	1A2A2A5J1-R	Less than 1.0
1A2A2A5J3-C	1A2A2A5J1-P	Less than 1.0
1A2A2A5J1-I	1A2A2A5J1-V	1000 ± 100
1A2A5A1J1-B	1A2A2A5J1-S	Greater than 100,000; 1000 ± 100 with BATTERY TEST depressed
1A2A5A1J1-E	1A2A2A4P1	Less than 1.0
1A2A2A4P2	1A2A2A4P3	Less than 1.0
1A2A2A4P4	1A2A2A4P3	169 ± 20
1A2A2A4P1	1A2A2A4P2	277 ± 30
1A2A2A3P1 shell	1A2A2A5J1-G	Less than 1.0
1A2A2A3P1 pin	1A2A2A5J1-E	Less than 1.0
1A2A2A5J1-A	1A2A2A5J1-B	20,000 ± 2000
1A2A2A5J1-A	1A2A2A5J1-C	Less than 100 to 20,000 ± 2000 with ZERO knob depressed and turned stop to stop; less than 100 at clockwise stop
1A2A2A4P2	1A2A5A1J1-B	Greater than 1000
1A2A2A5J3-C	1A2A2A5J1-W	Less than 1.0
1A2A5A1J1-E	1A2A5A1J1-B	Greater than 1000
1A2A2A5J1-N	1A2A2A5J1-O	Less than 1.0; greater than 1000 with ZERO knob depressed
1A2A5A1J1-E	1A2A2A4P1	Less than 1.0

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3.8.3 High temperature operation. When tested as specified in 4.4.4.17.3 the detector shall not false alarm (see 6.5.7) or malfunction (see 6.5.8). At the conclusion of the test, the detector shall have proper airflow at the air inlet and shall respond to the simulant solution as described in TM3-6665-225-12.

3.8.4 Low temperature operation. When tested as specified in 4.4.4.17.4 the detector shall not false alarm (see 6.5.7) or malfunction (see 6.5.8) and the detector operating input current shall fluctuate as shown in figure 1 and show no indication of case heater thermostat contact chatter. At the conclusion of the test, the detector shall have proper airflow and shall respond to the simulant solution as described in TM3-6665-225-12.

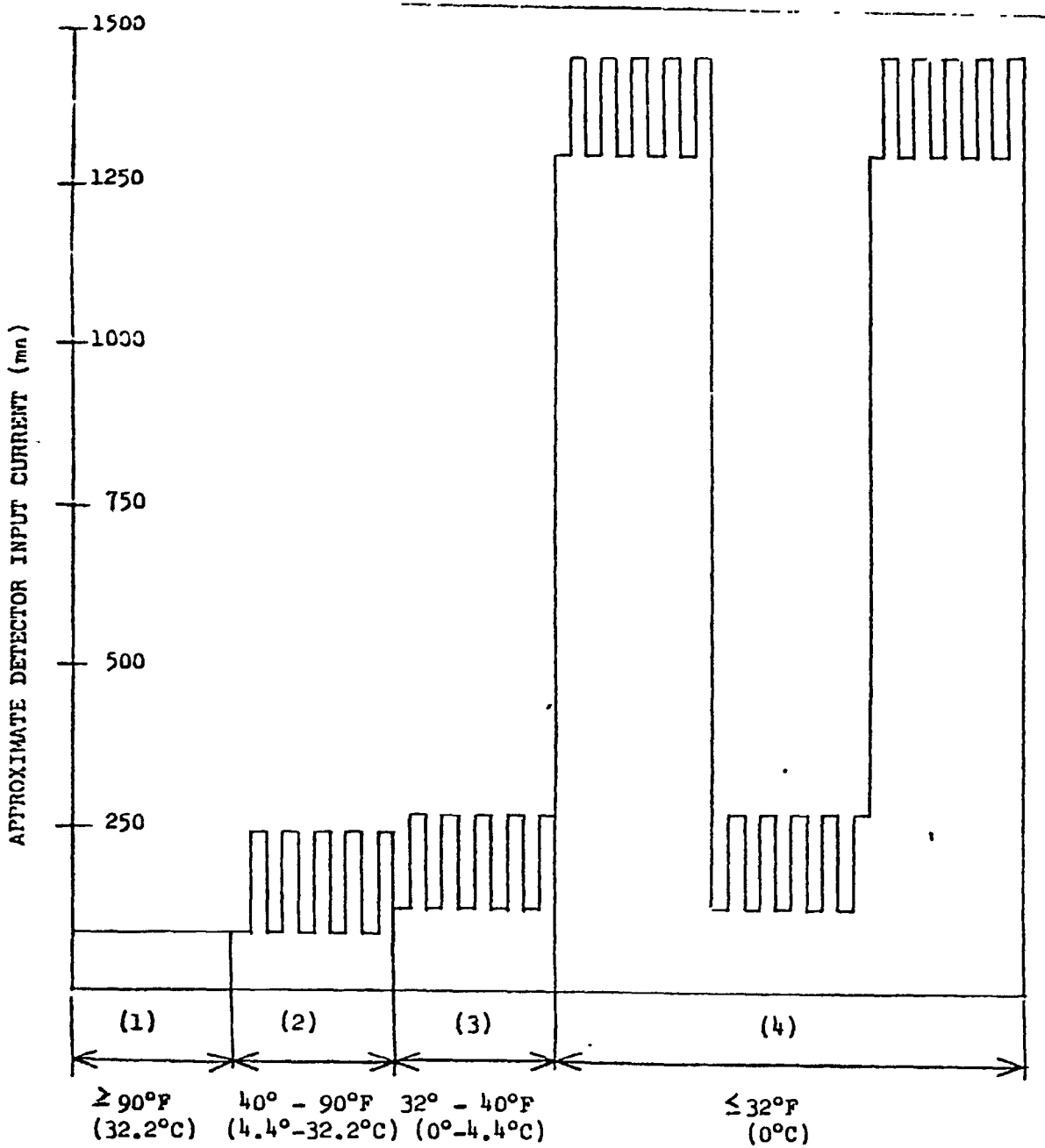
3.9 Sensitivity. When tested as specified in 4.4.4.18, the detector shall alarm to each of three successive exposures to a concentration (C) of 1.2 ± 0.2 gamma per liter (γ/l) of methanesulfonyl fluoride (MSF), Eastman grade, item 8368 (see 6.3) with a time (T) to alarm in minutes (min) such that the average CT is not greater than 0.80 γ min/l. Alarm shall be evidenced by the sounding of the detector's horn.

3.10 Operation at low temperature. When tested as specified in 4.3.3.1, the detector shall operate after an exposure of at least 3 hours duration to a temperature of -40° ± 5°F (-40° ± 2.78°C) while in a nonoperating condition, then operate without false alarm and with normal airflow through the detector (as defined in TM3-6665-225-12, "Operation of the Detector Unit Below 32°F") for at least 6 hours at a temperature of -40° ± 5°F, (-40° ± 2.78°C), and then meet the sensitivity requirement of 3.9 within 1 hour after removal from the low temperature environment.

3.11 Operation at high temperature. When tested as specified in 4.3.3.2, the detector shall operate after an exposure of at least 3 hours duration to a temperature of 105° ± 5°F (40.56° ± 2.78°C) while in a nonoperating condition, then operate without false alarm with normal airflow through the detector (as defined in TM3-6665-225-12, "Operation Under Usual Conditions") for at least 6 hours at a temperature of 105° ± 5°F, (40.56° ± 2.78°C) and then meet the sensitivity requirement of 3.9 within 1 hour after removal from the high temperature environment.

3.12 Operation during rain. When tested as specified in 4.3.3.3, the detector shall operate without false alarm during an exposure to rainfall at the rate of 4.0 ± 0.5 inch (10.16 ± 1.27 cm) per hour for at least 0.5 hour with the top panel of the detector up, and then with the detector canted 45° ± 5° and each of the four sides in turn exposed to rainfall at the rate of 4.0 ± 0.5 inch (10.16 ± 1.27 cm) per hour for at least 0.5 hour for a total rainfall exposure of at least 2.5 hours and then meeting the sensitivity requirements of 3.9 within 1 hour after removal of the detector from the rain environment. The accumulation of water within the case of the detector shall not be a cause for rejection.

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Operating Detector

(1) Less all heaters

(2) Plus cycling air inlet heater

(3) Plus air outlet heater and cycling air inlet heater

(4) Plus air outlet heater, cycling case bottom heater and cycling air inlet heater

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3.13 Operation after vibration. When tested as specified in 4.3.3.4, the detector shall operate without false alarm for at least 0.5 hour and then meet the sensitivity requirement of 3.9 after exposure to the vibration environment described in MIL-STD-810, method 514, procedure VIII, curve Y (6,000 miles), (9,656 kilometers). There shall be no physical deterioration of the unit after vibration.

3.14 Minimum detectable concentration. When tested as specified in 4.3.3.5, the detector shall alarm to each of three exposures to MSF at a concentration of 0.70 ± 0.10 γ/l within 5.0 minutes after the start of each exposure. Alarm shall be evidenced by the sounding of the detector's horn.

3.15 Preproduction and reliability qualification. Prior to the start of regular production, a preproduction sample and a reliability qualification sample of detectors shall be produced in accordance with this specification for examination and test (see 4.3).

3.16 Workmanship. The workmanship of the detector shall be in accordance with MIL-STD-252.

3.17 Reliability qualification.

3.17.1 Qualification phase and production acceptance phase of production reliability. When tested as specified in 4.3.3.6, the detector shall meet the following criteria for Mean Time Between Stoppages (MTBS) (see 6.5.3), Mean Time Between False Alarms (MTBFA) (see 6.5.4), and Mean Time Between Failures (MTBF) (see 6.5.5), while subjected to the test cycle described below and by figure 2.

Reliability criteria (hours)

	Upper test(1) (0_0)	Lower test(1) (0_1)
MTBS	351	234
MTBFA	750	500
MTBF	800	400

(1) As defined in MIL-STD-781

(a) Test cycle. Figure 2 shows a complete test cycle for the detector and the test cycle shall be continuously repeated for the duration of the test. Each temperature test shall be a total of twelve (12)

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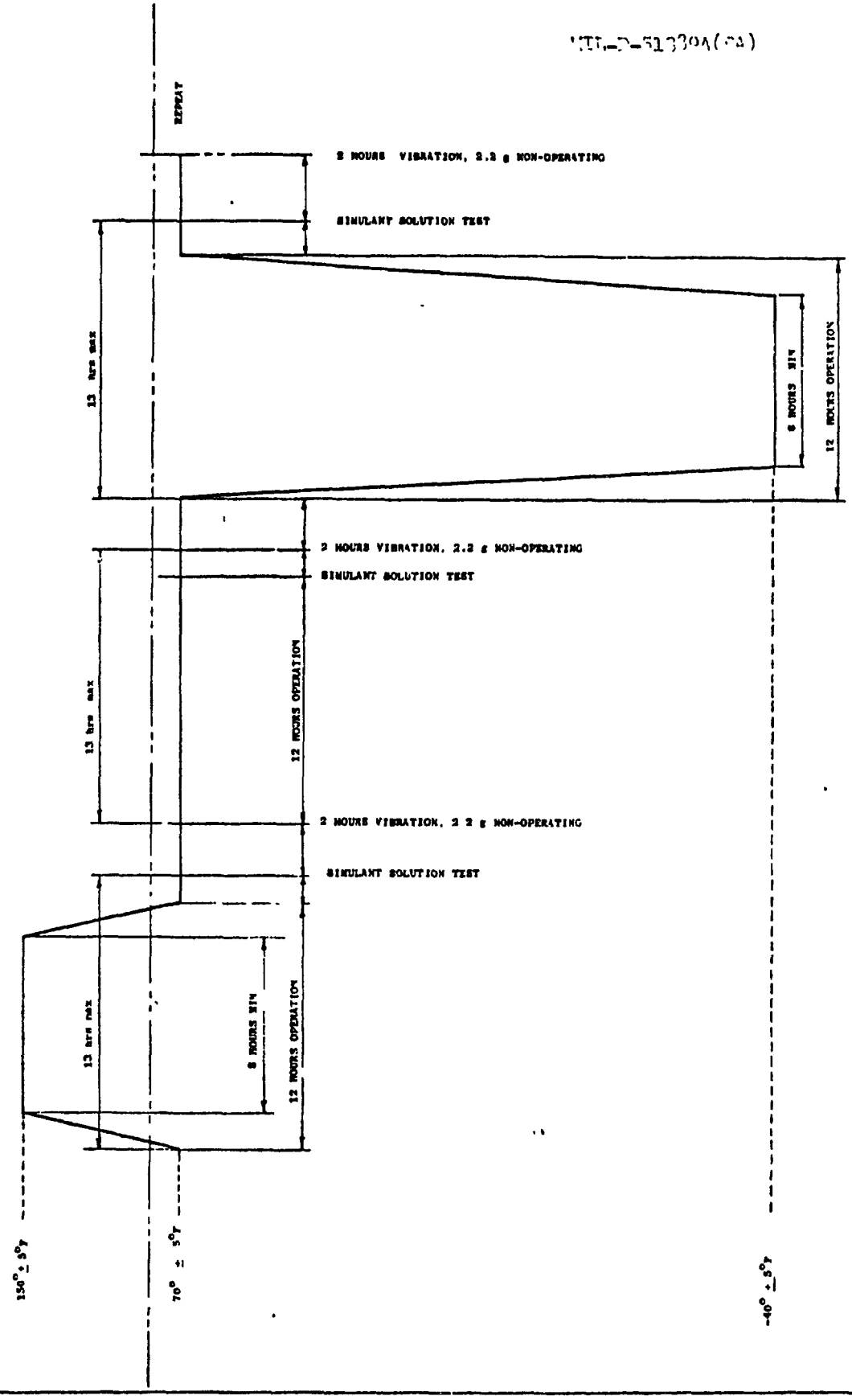


FIGURE 2 TEMPERATURE CYCLING FOR RELIABILITY TESTING

PART OF MIL-D-81393(EA)

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consecutive hours; a maximum of two (2) hours to attain the test temperature, a minimum of eight (8) hours at the test temperature, and a maximum of two (2) hours to return to 70° ± 5°F (21.1° ± 2.8°C). The detector response test shall be conducted within one (1) hour of completion of each temperature test and the duration of the vibration test shall be at least two (2) hours at the intervals shown on figure 2. The detector shall be operational during each temperature test and each detector response test and shall be nonoperational during each vibration test.

(b) Temperature test. Test temperatures shall be -40° ± 5°F (-40° ± 2.8°C), 70° ± 5°F (21.1° ± 2.8°C), and 105° ± 5°F (40.56 ± 2.8°C) as shown in figure 2.

(c) Detector response. The detector shall respond to the simulant solution as described in TM 3-6665-225-12 or shall meet the requirement for minimum detectable concentration as specified in 3.14.

(d) Vibration. 2.2 G ± 10% peak acceleration value at a nonresonant frequency between 20 and 60 Hertz in the vertical plane as measured at the equipment mounting points on the vibration table.

(e) Input voltage cycling. The input voltage cycling shall not be performed.

(f) Failure rate predictions. A failure rate prediction is not required.

(g) Thermal survey. A thermal survey is not required.

(h) Operation, maintenance, and repairs. The detector shall be operated, maintained, and repaired in accordance with the procedures described in TM3-6665-225-12.

(i) Detector operating parameters. The detector input voltage and current shall be recorded hourly. The detector shall be checked before and after each 12 hour temperature test for proper airflow.

(j) Detector response. At the conclusion of each temperature test, the detector shall respond to the simulant solution as described in TM3-6665-225-12 or shall meet the requirement for minimum detectable concentration as specified in 3.14.

(k) Reliability test reports. Reliability test reports in accordance with MIL-STD-781 are required except that the reliability design data and correlation of test results in the final report are not required.

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3.17.2 Definitions.

(a) Mean time between stoppages (MTBS). Mean operating time between the occurrence of all stoppages. A stoppage is defined as the occurrence of a false alarm (see 3.17.2(c)), malfunction, component failure, or any condition, not caused by operator error, test equipment or facility failure, which renders the system incapable of detecting or alarming to the presence of the specified concentration of MSF within the specified time. (Failures which cause false alarms are not charged as two stoppages.)

(b) Mean time between failures (MTBF). Mean operating time between the occurrence of all independent failures. A failure is defined as the occurrence of a malfunction, component failure, or loss of MSF detection or warning capability that is not:

(1) Caused by operator error
(2) Caused by test equipment or facility failure
(3) Readily correctable by the operator
(4) Readily repairable by the replacement of a "plug-in" module.

(c) Mean time between false alarm (MTBFA). Mean operating time between the occurrence of all false alarms that are not due to operator error, facility or test equipment failure. A false alarm is defined as occurring when the horn sounds without the presence of MSF unless the alarm was human induced, i.e., purposely sounded for testing purposes.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) Preproduction and reliability qualification inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction and reliability qualification inspection.

4.3.1 Sample.

4.3.1.1 Preproduction. A preproduction sample of eight detectors shall be manufactured using the same methods, materials, equipment, and processes as will be used during regular production.

4.3.1.2 Reliability qualification. A reliability qualification sample of 22 detectors shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production. The provisions of MIL-STD-781, Preconditioning burn-in, shall apply.

4.3.2 Inspection procedures for examination and test.

4.3.2.1 Preproduction. The preproduction sample of detectors shall be examined in accordance with the classification of defects 4.4.3.3 and tested in accordance with 4.4.4.1 through 4.4.4.18 and 4.3.3.1 through 4.3.3.5 for conformance to the requirements of 3.3 through 3.14.

4.3.2.2 Reliability qualification. The reliability qualification sample of detectors shall be examined in accordance with the classification of defects 4.4.3.3 and tested in accordance with 4.4.4.1 through 4.4.4.18 and 4.3.3.6 for conformance to the requirements of 3.3 through 3.9 and 3.17.

4.3.3 Tests. Tests shall be conducted as follows:

4.3.3.1 Operation at low temperature. Empty the reservoir of the detector to be tested, and reinstall the empty reservoir. Remove the air filter. Do not reinstall the case bottom. Place the detector and the case bottom into a temperature test chamber already at the temperature required by 3.10 and permit it to remain undisturbed at that temperature for at least 3 hours. Reinstall the case bottom and then start the detector in accordance with the instructions for operation below 32°F (0°C) contained in TM3-6665-225-12 except that:

(a) Bulk powder conforming to item no. 2 of Drawing C5-15-5599 and buffer solution conforming to Drawing C5-15-5479 may be used in servicing the detector.

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(b) Any suitable power source may be used in lieu of those specified in the technical manual. If an alternate power source is used, be careful to observe the correct polarity when connecting the power source to the detector.

(c) The procedure described in the technical manual for testing the sensitivity of the detector with a simulant should be omitted.

After normal operation has been established, permit the detector to operate undisturbed at the temperature specified in 3.10 for at least 6 hours. Monitor the detector for false alarm during this period of operation. Just prior to the end of the period of operation, check the air flow at the detector's air inlet in the same manner as the check was performed when the detector was being started. Then, remove the detector from the temperature test chamber. Subject the detector to the sensitivity test described in 4.4.4.18 except that the oxime solution shall not be changed, i.e., the oxime solution used during the exposure to the low temperature environment shall be used during the sensitivity test. A new air filter should be installed for the test. The test shall be completed within 1 hour after the detector was removed from the low temperature environment, and the sensitivity shall be as required by 3.9. Evaluate the results of the tests and observations against the requirements of 3.10.

4.3.3.2 Operation at high temperature. Empty the reservoir of the detector to be tested and reinstall the empty reservoir. Remove the air filter. Do not reinstall the case bottom. Place the detector and the case bottom into a temperature test chamber already at the temperature required by 3.11 and permit it to remain undisturbed at that temperature for at least 3 hours. Reinstall the case bottom and then start the detector in accordance with the instructions for normal operation contained in TM3-6665-225-12 except that:

(a) Bulk powder conforming to item no. 2 of Drawing C5-15-5599 and buffer solution conforming to Drawing C5-15-5479 may be used in servicing the detector.

(b) Any suitable power source may be used in lieu of those specified in the technical manual. If an alternate power source is used, be careful to observe the correct polarity when connecting the power source to the detector.

(c) The procedure described in the technical manual for testing the sensitivity of the detector with a simulant should be omitted.

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After normal operation has been established, permit the detector to operate undisturbed at the temperature specified in 3.11 for at least 6 hours. Monitor the detector for false alarm during this period of operation. Just prior to the end of the period of operation, check the airflow at the detector's air inlet in the same manner as the check was performed when the detector was being started. Then remove the detector from the temperature test chamber. Subject the detector to the sensitivity test described in 4.4.4.18 except that the oxime solution shall not be changed, i.e., the oxime solution used during the exposure to the high temperature environment shall be used during the sensitivity test. A new air filter should be installed for the test. The test shall be completed within 1 hour after the detector was removed from the high temperature environment, and the sensitivity shall be as required by 3.9. Evaluate the results of the tests and observations against the requirements of 3.11.

4.3.3.3 Operation during rain. Start the detector in accordance with the instructions for operation in rain contained in TM3-6665-225-12 except that:

(a) Bulk powder conforming to item no. 2 of Drawing C5-15-5599 and the buffer solution conforming to Drawing C5-15-5479 may be used in servicing the detector.

(b) Any suitable power source may be used in lieu of those specified in the technical manual. If an alternate power source is used, be careful to observe the correct polarity when connecting the power source to the detector.

(c) The procedure described in the technical manual for testing the sensitivity of the detector with a simulant should be omitted.

After normal operation has been established place the detector, with the top panel up, into a test chamber previously adjusted to produce rainfall at the rate required by 3.12. Turn on the rain for the period of time required by 3.12, and monitor the detector for false alarm during the rainfall period. Then, cant the detector 45° ± 5°. The rain may be interrupted for the period of time necessary to enter the test chamber and reposition the detector when repositioning the detector is required by this test procedure. Subject the detector to another period of the rainfall required by 3.12, and monitor the detector for false alarm during the rainfall period. Then reposition the detector such that it is still canted 45° ± 5° but another side is exposed to the rain. Subject the detector to another period of rainfall and monitor the detector for false alarm during the rainfall period. Continue this procedure until all four sides of the detector as well as the top panel have been exposed to the rain. Then, remove the detector from the test chamber

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and subject the detector to the sensitivity test described in 4.4.4.18 except that the oxime solution shall not be changed, i.e., the oxime solution used during the exposure to the rain environment shall be used during the sensitivity test. A new air filter shall be installed for the test. The test shall be completed within one hour after the detector was removed from the rain environment, and the sensitivity shall be as required by 3.9. Evaluate the results of the tests and observations against the requirements of 3.12.

4.3.3.4 Operation after vibration. Empty the reservoir of the detector to be tested, and reinstall the empty reservoir. Examine the detector to establish that the pump unit (Drawing D5-15-4600), the electronics module (Drawing D5-15-4550) and the cell (Drawing D5-15-4500 or D5-15-5610) are properly installed and secured. Reinstall the case bottom. Install the covers onto the electrical connector and the air outlet. Close the air inlet. Examine the rain shield and the flowmeter to establish that these components are properly installed and secured. Mount the detector onto a vibration exciter. The chains securing the electrical connector cover and the filter port cover to the detector may be taped to the detector to reduce noise during the test. Subject the detector to the vibration environment required by 3.13. Remove the detector from the vibration exciter. Start the detector in accordance with the instructions for normal operation contained in TM3-6665-225-12 except that:

(a) Bulk powder conforming to item no. 2 of Drawing C5-15-5599 and buffer solution conforming to Drawing C5-15-5479 may be used in servicing the detector.

(b) Any suitable power source may be used in lieu of those specified in the technical manual. If an alternate power source is used, be careful to observe the correct polarity when connecting the power source to the detector.

(c) The procedure described in the technical manual for testing the sensitivity of the detector with a simulant should be omitted.

After normal operation has been established, permit the detector to operate undisturbed for at least 0.5 hour. Monitor the detector for false alarm during this period of operation. Then, subject the detector to the sensitivity test described in 4.4.4.18 except that the oxime solution shall not be changed, i.e., the oxime solution used during the period of operation shall be used during the agent sensitivity test. A new air filter may be installed for the test. The sensitivity shall be as required by 3.9. Evaluate the results of the tests and observations against the requirements of 3.13. Examine the unit for physical deterioration.

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4.3.3.5 Minimum detectable concentration. Subject the detector unit to the concentrations specified in 3.14 using the test procedures as specified in 4.4.4.18.

4.3.3.6 Reliability qualification. The detectors shall be tested as specified herein with "accept" and "reject" criteria in accordance with the following test plans of MIL-STD-781:

MTBS - Test plan IC

MTBFA - Test plan IIC

MTBF - Test plan IIIC

Continue testing until either an "accept" decision is reached for all three criteria (MTBS, MTBFA, and MTBF) or a "reject" decision is reached on one of the three criteria (MTBS, MTBFA, or MTBF).

4.3.3.6.1 Test planning. A reliability test plan document and reliability test procedures shall be prepared in accordance with Reliability test planning, Reliability test plan document, and Reliability test procedures of MIL-STD-781.

4.3.3.6.2 Operation, maintenance, and repairs. The detector shall be operated, maintained, and repaired in accordance with the procedures described in TM3-6665-225-12 and TM3-6665-302-34, except that any suitable power source supply 28.0 ± 9.0 vdc may be used in lieu of those specified in the technical manual. If an alternate power source is used, be careful to observe the correct polarity when connecting the power source to the detector. The provisions for Verifying repair and Preventive maintenance of MIL-STD-781 shall apply.

4.3.3.6.3 Test facilities. Test facilities used for the reliability qualification test shall comply with the provisions of Test facilities of MIL-STD-781.

4.3.3.6.4 Failure actions. On the occassion of a stoppage (see 6.5.6) the provisions of Failure actions of MIL-STD-781 shall apply.

4.3.3.6.5 Corrective action. Corrective action for all stoppages (see 6.5.6) shall be accomplished in accordance with the provisions of Corrective action of MIL-STD-781.

4.3.3.6.6 Test records. Test records shall be maintained in accordance with Test records of MIL-STD-781.

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4.3.3.6.7 Restoration of tested equipment. Upon completion of the reliability qualification test the detectors shall be refurbished in accordance with Restoration of tested equipments of MIL-STD-781.

4.3.3.6.8 Test procedures.

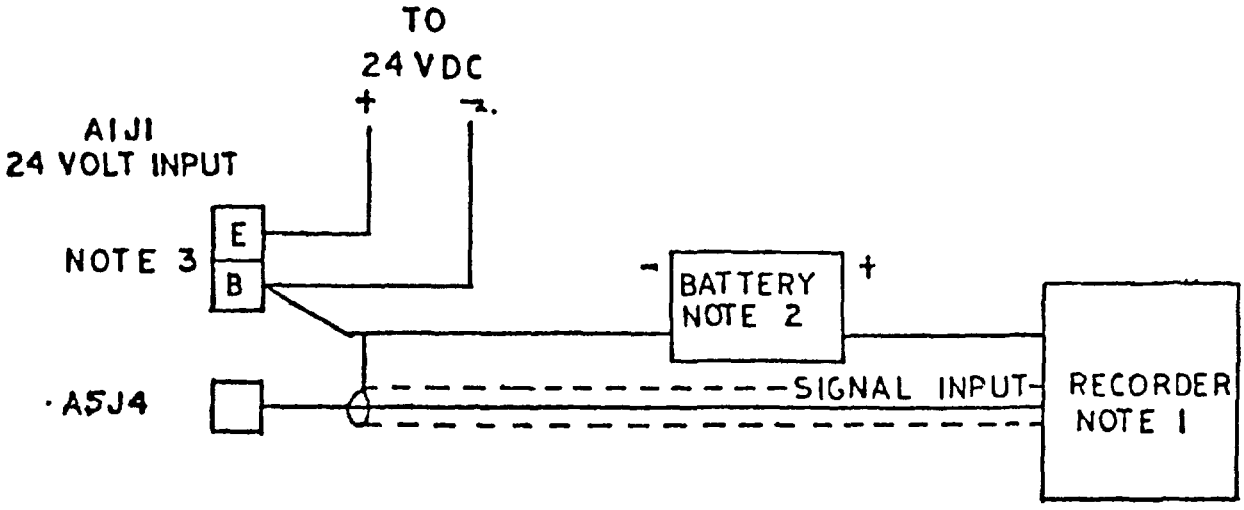
4.3.3.6.8.1 High temperature operation (105° ± 5°F) (40.56° ± 2.78°C). Start the detector in accordance with the instructions for normal operation contained in TM3-6665-225-12. Attach all required test instrumentation to continuously monitor the detector amplifier output (A5J4) with a suitable strip chart recorder as shown in figure 3. Attach all required test instrumentation to monitor the detector input current with a suitable strip chart recorder a minimum of five consecutive minutes during each hour of operation.

After normal operation has been established, place the detector in an environmental test chamber and allow the detector to operate undisturbed for a minimum of 12 consecutive hours; a minimum of eight consecutive operating hours shall be at a test temperature of 105° ± 5°F (40.56° ± 2.8°C). It is recommended that the operating detector not be subjected to a rate of temperature change in excess of 2.0°F (1.1°C) per every five minutes. Within one hour following completion of the test, the detector shall be checked for proper airflow at the air inlet in the same manner as the check was performed when the detector was being started and shall be checked for response to the simulant solution as specified in TM3-6665-225-12. The detector shall not be shut off between the period of completion of the 12 hour temperature test and the completion of the airflow and simulant solution response check and oxime solution shall not be changed.

4.3.3.6.8.2 Temperature operation (70° ± 5°F) (21.1° ± 2.8°C). Start the detector in accordance with the instructions for normal operation contained in TM3-6665-225-12. Attach all required test instrumentation to continuously monitor the detector amplifier output (A5J4) with a suitable strip chart recorder as shown in figure 3. Attach all required test instrumentation to monitor the detector input current with a suitable strip chart recorder a minimum of five consecutive minutes during each hour of operation.

After normal operation has been established, place the detector in an environmental test chamber and allow the detector to operate undisturbed for a minimum of 12 consecutive hours; a minimum of eight consecutive operating hours shall be at a test temperature of 70° ± 5°F (21.1° ± 2.8°C). It is recommended that the operating detector not be subjected to a rate of temperature change in excess of 2.0°F (1.1°C) per every five minutes. Within one hour following completion of the test, the detector shall be

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Notes:

1. Recorder characteristics:

   (a) Input impedance of 100,000 ohms, minimum
   (b) Input floating, i.e., not grounded

2. Battery is a 8.5 ± 1.0 volt mercury battery. This battery not necessary if the recorder balance control has sufficient range to permit zeroing recorder.

3. Connection between the minus of the 24 volt supply and the minus of the mercury battery must be made at the connector pin.

Figure 3. Monitoring cable and recorder.

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checked for proper airflow at the air inlet in the same manner as the check was performed when the detector was being started and shall be checked for response to the simulant solution as specified in TM3-6665-225-12. The detector shall not be shut off between the period of completion of the 12 hour temperature test and the completion of the airflow and simulant solution response check and oxime solution shall not be changed.

4.3.3.6.8.3 Low temperature operation (-40° ± 5°F) (-40° ± 2.8°C). Start the detector in accordance with the instructions for normal operation contained in TM3-6665-225-12. Attach all required test instrumentation to continuously monitor the detector amplifier output (A5J4) with a suitable strip chart recorder as shown in figure 3. Attach all required test instrumentation to monitor the detector input current with a suitable strip chart recorder a minimum of five consecutive minutes during each hour of operation.

After normal operation has been established, place the detector in an environmental test chamber and allow the detector to operate undisturbed for a minimum of 12 consecutive hours; a minimum of eight consecutive operating hours shall be at a test temperature of -40° ± 5°F (-40° ± 2.8°C). It is recommended that the operating detector not be subjected to a rate of temperature change in excess of 2.0°F (1.1°C) per every five minutes. Within one hour following completion of the test, the detector shall be checked for proper airflow at the air inlet in the same manner as the check was performed when the detector was being started and shall be checked for response to the simulant solution as specified in TM3-6665-225-12. The detector shall not be shut off between the period of completion of the 12 hour temperature test and the completion of the airflow and simulant solution response check and oxime solution shall not be changed.

4.3.3.6.8.4 Vibration. Empty the reservoir of the detector to be tested and reinstall the empty reservoir. Remove the air filter from the air filter slot of the detector and install and tighten the air filter plug. Examine the detector to establish that the pump unit, the electronics module and the cell are properly installed and secured. Install the covers onto the electrical connector and the air outlet. Close the air inlet and screw the plug into the air inlet assembly. Examine the rain shield and the flowmeter to establish that these components are properly installed and secured. Mount the detector onto a vibration table in the vertical position. Pull up the hand crank to the storage position. Subject the detector to 2.2G + 10% peak acceleration value at a nonresonant frequency between 20 and 60 Hertz in the vertical plane as measured at the equipment mounting points on the vibration table. The duration of vibration shall be at least two hours. The vibration transducer shall be "on" and monitored during vibration. Upon completion of the two hour vibration period, remove the detector from the vibration table and visually examine the unit for physical deterioration.

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4.3.4 Acceptance/rejection criteria. The preproduction sample and reliability qualification sample of detectors shall comply with the examination and tests specified in 4.3.2.1 and 4.3.2.2 to be acceptable. The provisions of Determination of compliance of MIL-STD-781 shall apply for the reliability qualification sample of detectors. The contractor shall obtain written approval from the contracting officer before proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the detectors offered for acceptance at one time, which have been produced by one manufacturer, at one plant, under essentially the same manufacturing conditions.

4.4.2 Sampling. Sampling for examination shall be conducted in accordance with MIL-STD-105.

4.4.3 Inspection procedures.

4.4.3.1 For examination. Sample detectors and the preparation for delivery shall be examined in accordance with the classification of defects 4.4.3.3 and MIL-STD-105.

4.4.3.2 For tests. Each detector unit in the lot shall be tested in accordance with 4.4.4.1 through 4.4.4.18. Any detector found defective shall be rejected. After completion of the tests, the detector unit shall be flushed as specified in 4.4.4.20.

4.4.3.3 Classification of defects.

(a) Detector unit, chemical agent automatic alarm (Drawing D5-15-4400).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Workmanship (applicable defects listed as Major in MIL-STD-252, (see 3.16)

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Minor:	AQL 4.0 percent defective	
201		Workmanship (applicable defects listed as Minor in MIL-STD-252, see 3.16)
	AQL 6.5 percent defective	
202		Workmanship (applicable defects listed as Control in MIL-STD-252, see 3.16)

(b) Packaging (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Packaging component missing, incorrect, or incorrectly assembled
102	Packaging component damaged
103	Markings missing, incorrect, or illegible

4.4.4 Tests. Tests shall be conducted and the detector units flushed as follows: The tests may be performed in any convenient order at a temperature of 75° ± 15°F (23.9° ± 8.3°C), except that 4.4.4.17 and 4.4.4.18 shall be the last tests performed, and 4.4.4.18 shall be performed after 4.4.4.17.

4.4.4.1 Airflow at air inlet. Determine the airflow utilizing the Q200 (D136-45-245) in accordance with the instruction manual.

4.4.4.2 Resistance to airflow. Determine the resistance to airflow utilizing the Q200 in accordance with the instruction manual.

4.4.4.3 Air leakage at the detector's air filter seal. Determine the leakage utilizing the Q200 in accordance with the instruction manual.

4.4.4.4 Air leakage of the combined air and liquid system. Determine the air leakage utilizing the Q200 in accordance with the instruction manual.

4.4.4.5 Air leakage of the liquid line. Determine the air leakage utilizing the Q200 in accordance with the instruction manual.

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4.4.4.6 Chassis heater circuit current. Determine the chassis heater circuit current utilizing the Q200 in accordance with the instruction manual.

4.4.4.7 Interior thermostat operation. Test interior thermostat operation utilizing the Q200 in accordance with the instruction manual.

4.4.4.8 Outlet heater circuit current. Determine the outlet heater circuit current utilizing the Q200 in accordance with the instruction manual.

4.4.4.9 Inlet heater current and operating temperature. Determine the inlet heater current utilizing the Q200 in accordance with the instruction manual.

4.4.4.10 Simulated alarm. Test the detector's response to a simulated alarm utilizing the Q200 in accordance with the instruction manual.

4.4.4.11 Meter zero adjustment circuit. Test the meter zero adjustment circuit utilizing the Q200 in accordance with the instruction manual.

4.4.4.12 Meter accuracy. Test the meter accuracy utilizing the Q200 in accordance with the instruction manual.

4.4.4.13 Detector operating current. Determine the detector operating current utilizing the Q200 in accordance with the instruction manual.

4.4.4.14 Resistances. Determine the resistances utilizing the Q200 in accordance with the instruction manual.

4.4.4.15 Fast priming mechanism. Connect a source of 24.0 ± 0.5 vdc to the detector's 24 VDC INPUT and start the detector's pump unit. Place the shaft of the PRIME crank in its center position. While observing the shouldered pin (Drawing C5-15-4648), pull the shaft of the PRIME crank into its uppermost position. The shouldered pin must be displaced as required by 3.6(a). While observing the shouldered pin and the roller (Drawing B5-15-4533) in the bell crank assembly, push the shaft of the PRIME crank into its center position. The relationship between the roller and the shouldered pin must change as required by 3.6 (b). While observing the end of the PRIME crank shaft, push the shaft into its lowermost position and rotate the crank. The end of the shaft must contact the fin housing (Drawing C5-15-4608), the pin in the end of the shaft must engage the fast prime fin (Drawing C5-15-4618) and rotate the fin housing as required by 3.6 (c). Release the PRIME crank. The shaft of the PRIME crank must return to its center position as required by 3.6 (c).

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4.4.4.16 Detector leakage. Leakage shall be determined by removing the pump unit and cell from the detector and the rainshield and air outlet cap installed. Apply the specified air pressure through the rainshield and submerge in water for the required time. Evidence of leakage shall be continuous bubbles.

4.4.4.17 All equipment screening test. Start the detector in accordance with the instructions for normal operation contained in TM3-6665-225-12 except that:

4.4.4.17.1 Vibration. Empty the reservoir of the detector to be tested and reinstall the empty reservoir. Remove the air filter from the air filter slot of the detector and install and tighten the air filter plug. Examine the detector to establish that the pump unit, the electronics module and the cell are properly installed and secured. Install the covers onto the electrical connector and the air outlet. Close the air inlet and screw the plug into the air inlet assembly. Examine the rain shield and the flowmeter to establish that these components are properly installed and secured. Mount the detector onto a vibration table in the vertical position. Pull up the hand crank to the storage position. Subject the detector to 2.2G+ 10% peak acceleration value at a nonresonant frequency between 20 and 60 Hertz (as determined in 4.3.3.4) in the vertical plane as measured at the equipment mounting points on the vibration table. The duration of vibration shall be at least two hours. The vibration transducer shall be "on" and monitored during vibration. Upon completion of the two hour vibration period, remove the detector from the vibration table and visually examine the unit for physical deterioration as required by 3.8.1. Upon successful completion of the requirement of this test the detector shall be tested as specified in 4.4.4.17.2.

4.4.4.17.2 Stability.

(a) Start the detector in accordance with the instructions for normal operation contained in TM3-6665-225-12 except that:

(1) Bulk powder conforming to item no. 2 of Drawing C5-15-5599 and buffer solution conforming to Drawing C5-15-5479 may be used in servicing the detector.

(2) Any suitable power source supplying 28.0 $\pm$ 9.0 vdc may be used in lieu of those specified in the technical manual. If an alternate power source is used, be careful to observe the correct polarity when connecting the power source to the detector.

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(3) The procedure described in the technical manual for testing the sensitivity of the detector with a simulant may be omitted.

(b) Monitor the detector's amplifier output (A5J4) with a suitable strip chart recorder as shown on figure 3. After at least 10 minutes of normal operation, rezero the detector. Allow the detector to operate undisturbed for at least 12 hours but not more than 24 hours. Within one hour following the completion of the test, the detectors shall be checked for proper airflow at the air inlet in the same manner as the check was performed when the detector was being started as specified in TM3-6665-225-12.

(c) After checking airflow and still within the one hour following completion of the initial stability test, test and set, if necessary, the air inlet heater temperature as follows:

(1) Disconnect input power and within 2 minutes, perform steps (2), (3) & (4).

(2) Open filter cap and remove filter paddle assembly and save in a clean container.

(3) Insert an instrumented temperature monitoring paddle and cap assembly equivalent to D136-45-256 used in the Q200 or D5-15-5534 used in the M74 test set, which has been equilibrated to 92° ± 5°F (33.3° ± 2.8°C), into the detector filter port.

(4) Reconnect input power.

(5) Monitor and adjust, if necessary, the inlet heater temperature to meet the requirements of 3.4.4 using procedures of figure 3-11 of TM3-6665-302-34 and a visual indicator calibrated to represent the actual temperature limits.

(6) Disconnect input power and within 2 minutes perform steps (7), (8), and (9).

(7) Remove instrumented filter paddle.

(8) Re-install filter paddle saved from step (2).

(9) Reconnect input power.

(d) Within one hour following completion of the stability test, with a stable amplifier output, the detector shall be checked for response to the simulant solution as specified in TM3-6665-225-12. The

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detector shall not be shut off between the period of completion of the 12 hour ambient temperature test [except the initial ambient stability test where the inlet heater temperature test and setting is performed per above para.(c)] and the completion of the airflow and simulant solution response check. The oxime solution shall not be changed.

(e) Evaluate the recording of the detector performance and the results of the tests and observations against the requirements of 3.8.2. Upon successful completion of the requirements of this test, the detector shall be tested as specified in 4.4.4.17.3 and 4.4.4.17.4. Failed detectors shall be retested as specified in 4.4.4.17.1 and 4.4.4.17.2 except 4.4.4.17.2(c) following repair.

4.4.4.17.3 High temperature operation. Start the detector in accordance with the instructions for normal operation contained in TM3-6665-225-12 except that:

(a) Bulk powder conforming to item no. 2 of drawing C5-15-5599 and buffer solution conforming to Drawing C5-15-5479 may be used in servicing the detector.

(b) Any suitable power source supplying 28.0 ± 9 vdc may be used in lieu of those specified in the technical manual. If an alternate power source is used, be careful to observe the correct polarity when connecting the power source to the detector.

(c) The procedure described in the technical manual for testing the sensitivity of the detector with a simulant may be omitted.

After normal operation has been established, place the detector in an environmental test chamber and allow the detector to operate undisturbed for a minimum of 12 consecutive hours; a minimum of eight consecutive operating hours shall be at a test temperature of 105° ± 5°F (40.6° ± 2.8°C). It is recommended that the operating detector not be subjected to a rate of temperature change in excess of 2.0°F (1.1°C) per every five minutes. Within one hour following completion of the test, the detector shall be checked for proper airflow at the air inlet in the same manner as the check was performed when the detector was being started and shall be checked for response to the simulant solution as specified in TM3-6665-225-12. The detector shall not be shut off between the period of completion of the 12 hour temperature test and the completion of the airflow and simulant solution response check and oxime solution shall not be changed. Evaluate the results of tne tests and observations against the requirements of 3.8.3. Failed detectors shall be retested as specified in 4.4.4.17.1 and 4.4.4.17.2 except 4.4.4.17.2(c) following repair.

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4.4.4.17.4 Low temperature operation. Start the detector in accordance with the instructions for normal operation contained in TM3-6665-225-12 except that:

(a) Bulk powder conforming to item no. 2 of Drawing C5-15-5599 and buffer solution conforming to Drawing C5-15-5479 may be used in servicing the detector.

(b) Any suitable power source supplying 28.0 ± 9.0 vdc may be used in lieu of those specified in the technical manual. If an alternate power source is used, be careful to observe the correct polarity when connecting the power source to the detector.

(c) The procedure described in the technical manual for testing the sensitivity of the detector with a simulant may be omitted.

After normal operation has been established, place the detector in an environmental test chamber and allow the detector to operate undisturbed for a minimum of 12 consecutive hours, a minimum of eight consecutive hours shall be at a test temperature of -40° ± 5°F (-40° ± 2.8°C). It is recommended that the operating detector not be subjected to a rate of temperature change in excess of 6.0°F (3.33°C) per every five minutes. Monitor the detector input current with a suitable strip chart recorder a minimum of five consecutive minutes during each hour of operation. Within one hour following completion of the test, the detector shall be checked for proper airflow at the air inlet in the same manner as the check was performed when the detector was being started and shall be checked for response to the simulant solution as specified in TM3-6665-225-12. The detector shall not be shut off between the period of completion of the 12 hour temperature test and the completion of the airflow and simulant solution response check and the oxime solution shall not be changed. Evaluate the recording of the detector input current and the results of the tests and observations against the requirements of 3.8.4. Failed detectors shall be retested as specified in 4.4.4.17.1 and 4.4.4.17.2 except 4.4.4.17.2(c) following repair.

4.4.4.18 Sensitivity. Start the detector in accordance with the instructions for normal operation contained in TM3-6665-225-12 except that:

(a) Bulk powder conforming to item no. 2 of Drawing C5-15-5599 and buffer solution conforming to Drawing C5-15-5479 may be used in servicing the detector.

(b) Any suitable power source may be used in lieu of those specified in the technical manual. If an alternate power source is used, be careful to observe the correct polarity when connecting the power source to the detector.

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(c) The procedure described in the technical manual for testing the sensitivity of the detector with a simulant should be omitted.

Install a test adapter fitting in the air inlet of the detector to facilitate connecting the detector to the agent source. This MSF source shall be a Portable Gas Generator Q219 (E136-91-1520). Demonstrate that the airflow at the detector air inlet is 1.1 (+ 0.4 or - 0.0) liter per minute. After the detector has been in operation at least 5.0 minutes after zeroing, connect the detector to the MSF source and start a timer. When the detector horn sounds stop the timer and disconnect the detector from the MSF source. Depress the ZERO knob fully and hold it in the depressed position at least 5.0 seconds (it may be necessary to repeat this operation several times to restore the detector to operation). Record the elapsed time indicated by the timer. Allow the detector to operate at least 2.0 minutes after the meter pointer has returned within the zero zone. Zero the detector meter, and again expose the detector to the MSF within 5.0 minutes after zeroing. Repeat the procedure described above until 3 exposures to the MSF have been accomplished. Evaluate the data accumulated against the requirements of 3.9. Determine the concentration of the MSF coming from the dilution apparatus prior to running any sensitivity test and every 4 hours during the tests.

4.4.4.19 Flushing. Remove the pump unit (Drawing E5-15-4600) and the detector cell (Drawing D5-15-4500 or D5-15-5610) from the detector subassembly. Flush any oxime solution residues from the detector subassembly chassis and the pump unit as described below, and then reinstall a pump unit into the detector subassembly. Process the detector cell as required by Drawing D5-15-4500 or D5-15-5610.

(a) Detector subassembly chassis. Flush each liquid-carrying line and passage in the chassis, including the line between the chassis seal at the detector cell air-liquid outlet and the reservoir, with deionized water at a maximum pressure of 45 inches (114.3 cm) of water above atmospheric pressure. Then, blow-out each line and passage in the chassis with filtered ambient air at a maximum pressure of 45 inches of water above atmospheric pressure. When the air inlet system is blown out the direction of airflow must be from the detector cell interface toward the air inlet, the machine thread plug (Drawing C5-15-4385) must be off, and the air filter removed.

(b) Pump unit. Examine the air inlet and outlet ports of pump unit air pump for the presence of oxime solution residues. If such residues are found, clean the air pump by flushing with deionized water. The flushing water must be forced into the air inlet port of the air

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pump at a maximum pressure of 45 inches (114.3 cm) of water above atmospheric pressure. After flushing the air pump, blow it out with filtered ambient air at a maximum pressure of 45 inches of water above atmospheric pressure. The air must enter the air pump at the inlet port. Each pump unit liquid pump must be flushed with deionized water and then blown out with filtered ambient air, both at a maximum pressure of 45 inches of water above atmospheric pressure.

5. PACKAGING

5.1 Preservation and unit packing, level A. The detector unit shall be unit packed in accordance with Drawing D5-15-6567.

5.2 Packing. Level A or commercial (see 6.2).

5.2.1 Level A. The detector unit, unit packed as specified in 5.1, shall be packed in containers conforming to the following:

(a) Box, wood, class 2, style 4, grade B, type 1 load, weight of contents 0-50 pounds, of PPP-B-621.

(b) Box, cleated plywood, overseas type, style optional, grade B, type 1 load, weight of contents 0-100 pounds, of PPP-B-601. The inside dimensions of the boxes shall be 9-1/8 by 9-1/8 by 13-1/8 inches (23.18 by 23.18 by 33.34 cm) (+ 1/8 inch [0.32 cm) tolerance on each of the dimensions) length, width, and depth, respectively. The boxes shall be closed and strapped in accordance with the appendix of their respective specifications. Strapping and seals shall conform to QQ-S-781 [Strapping - type I (regular duty), finish B, grade 2, and Seals - type D, style I or IV, finish B, grade 2].

5.2.2 Commercial packing. A uniform quantity of detector units, unit packed as specified in 5.1, shall be prepared for shipment as specified in MIL-STD-1188.

5.3 Marking. In addition to any special marking required by the contract or order, unit packs (5.1) shall be marked in accordance with MIL-STD-129 (interior unit pack markings). Level A shipping containers (5.2.1) shall be marked in accordance with Drawing D5-15-6564. Commercial shipping containers (5.2.2) shall be marked in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The detector unit is intended to detect and warn of the presence of V and G agents.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preproduction.

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

(c) Level A or commercial packing required.

6.3 Test reagent. Methane sulfonyl fluoride, Eastman grade, item 8368, boiling point 123-12h° centigrade (253.4 - 255.2°F), may be purchased from Distillation Products Industries, Eastman Organic Chemicals Dept, Rochester, NY 14603.

6.4 International interest. Certain provisions of this specification are the subject of international standardization agreement QSTAG 251. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

6.5 Definitions. For the purpose of this specification, the following definitions shall apply:

6.5.1 Drift. A long term change (over one minute) in the level of the mean amplifier electrical output in either the plus or minus direction.

6.5.2 Noise and transients. Variations in amplifier electrical output of one minute or less in duration.

6.5.3 Mean time between stoppages (MTBS). Mean operating time between the occurrence of all stoppages (see 6.5.4).

6.5.4 Stoppage. The occurrence of a false alarm (see 6.5.7) or malfunction (see 6.5.8).

6.5.5 Mean time between false alarms (MTBFA). Mean operating time between the occurrence of all false alarms (see 6.5.7).

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6.5.6 Mean time between failures (MTBF). Mean operating time between the occurrence of all failures (see 6.5.9).

6.5.7 False alarm. The sounding of the detector horn without the presence of the test reagent, MSF, or the field simulant, BSC, unless the alarm was human induced (i.e., purposely sounded for testing purposes) and not due to operator error (see 6.5.10), test equipment or facility failure.

6.5.8 Malfunction. Any detector anomaly requiring corrective maintenance detector failure or inability to alarm to specified concentrations of the test reagent, MSF, or the field simulant, BSC, within required times; inability to operate the detector in accordance with TM3-6665-225-12. Malfunctions caused by operator error (see 6.5.10), test equipment or facility failure are non relevant (see 3.1.5.2, MIL-STD-781, Nonrelevant failure). Malfunctions which cause false alarms shall be considered as one stoppage.

6.5.9 Failure. A stoppage (see 6.5.4) that is not correctable at organizational or a lower maintenance level as defined in Appendix B, Maintenance Allocation Chart, TM3-6665-225-12.

6.5.10 Operator error. Operation of the detector in a manner not in conformance with TM3-6665-225-12.

6.6 Contract data requirements. Items of deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4.3.3.6.1	Plan, Reliability Test	DI-R-7033
4.3.3.6.1	Procedures, Reliability Test	DI-R-7035
4.3.3.6.5	Plan, Corrective Action	DI-R-7038

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